

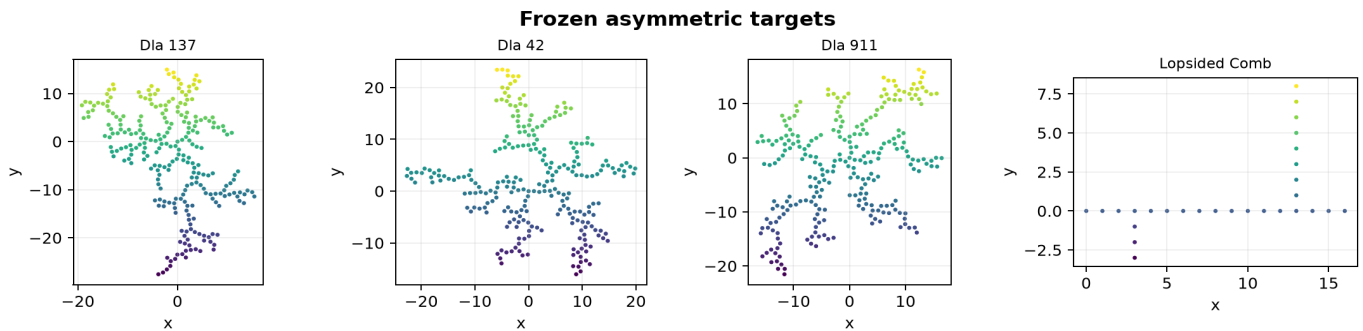
Asymmetric-target finite-boundary validation

A preregistered benchmark of exact Poisson, uniform restart, one-shot centering, and the four dynamic approaches

Executive result

The experiment creates a non-circular target geometry while keeping the death boundary circular, so the comparison is non-circular in morphology without changing the theorem's boundary assumptions. Exact Poisson return remains the reference law. As the killing radius grows, uniform restart and one-shot centering approach that reference; centering is the certified approximation that can remove the dominant finite-boundary bias while preserving a practical restart implementation.

The pilot completed 180 validated records and triggered doubled evaluation counts for 9 of 9 radii. Production therefore used 50,000 evaluation attachments per policy and 8,192 search probes per centered replication.

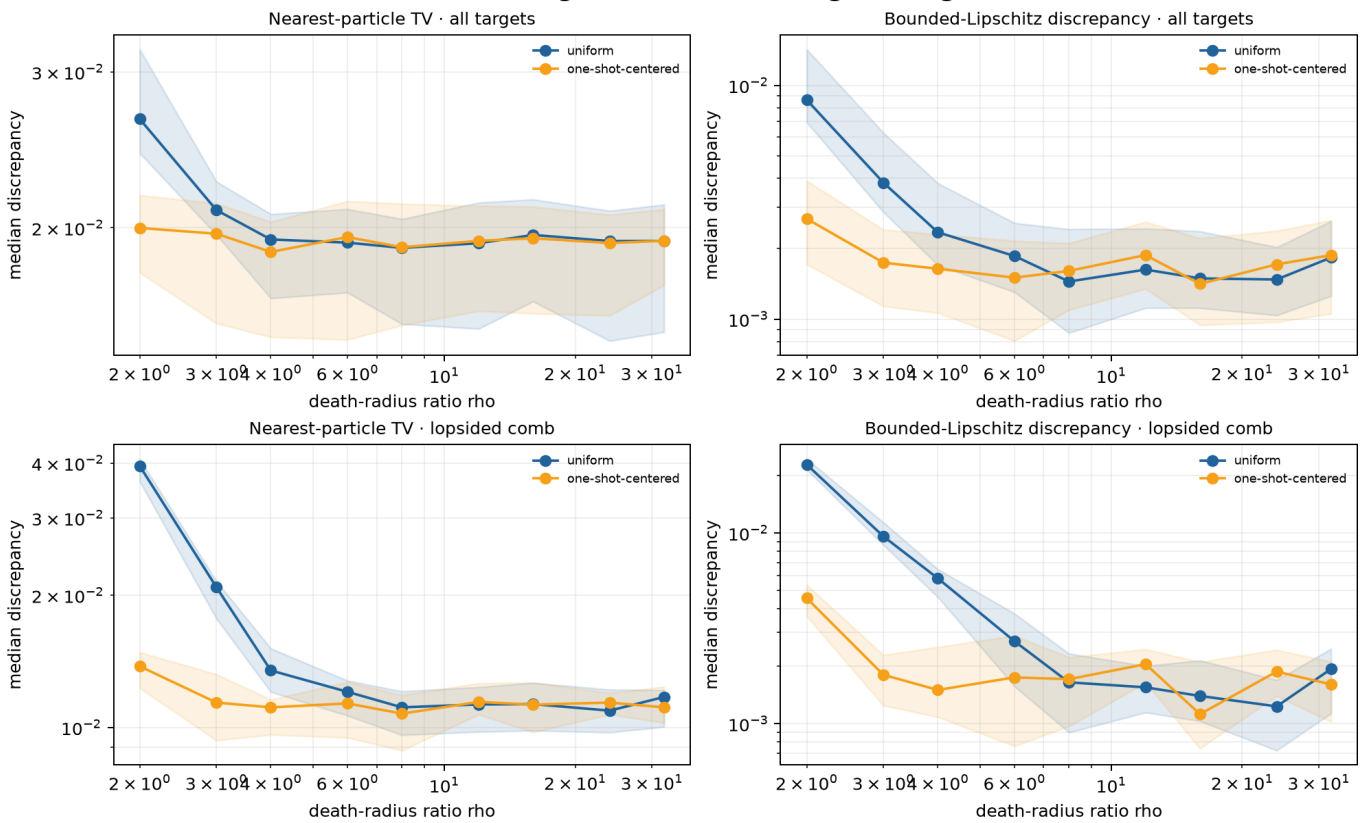


Scope and interpretation

The reported TV values are empirical diagnostics on fixed finite partitions (nearest-particle IDs and angular histograms), not continuum total-variation proofs. The target suite is one deterministic lopsided comb plus three exact-seed DLA targets (256 particles; seeds 42, 137, 911). All static comparisons use the same target, circular boundary construction, and independent deterministic streams.

1. Geometry and experimental contract

Static frozen-target error as the killing radius grows

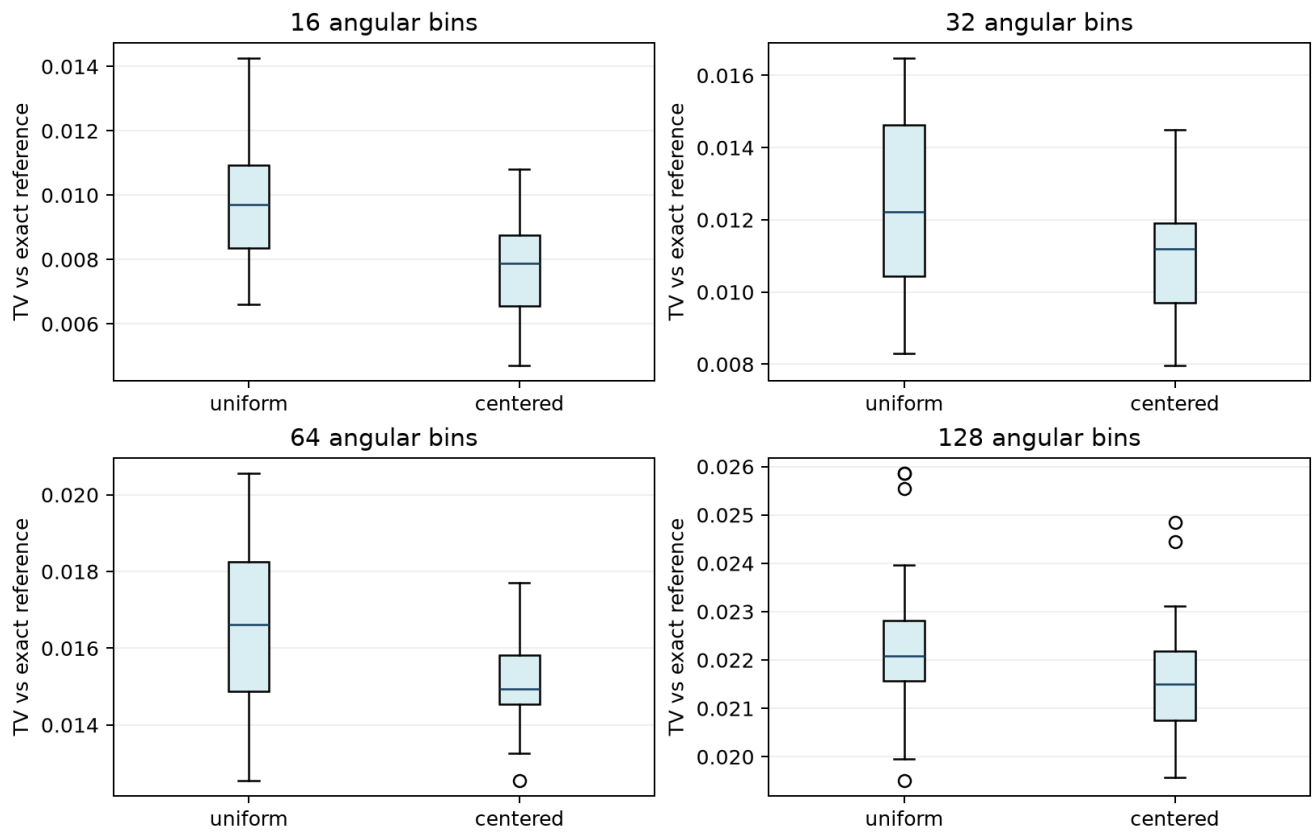


The top row pools all targets; the bottom row isolates the deliberately lopsided comb, where the centering advantage is clearest at small killing radii. Lines are medians across replications and shaded bands are interquartile ranges. The preregistered slope window is ρ in $\{3, 4, 6, 8, 12, 16\}$; $\rho = 2, 24, 32$ are diagnostics outside the fit window.

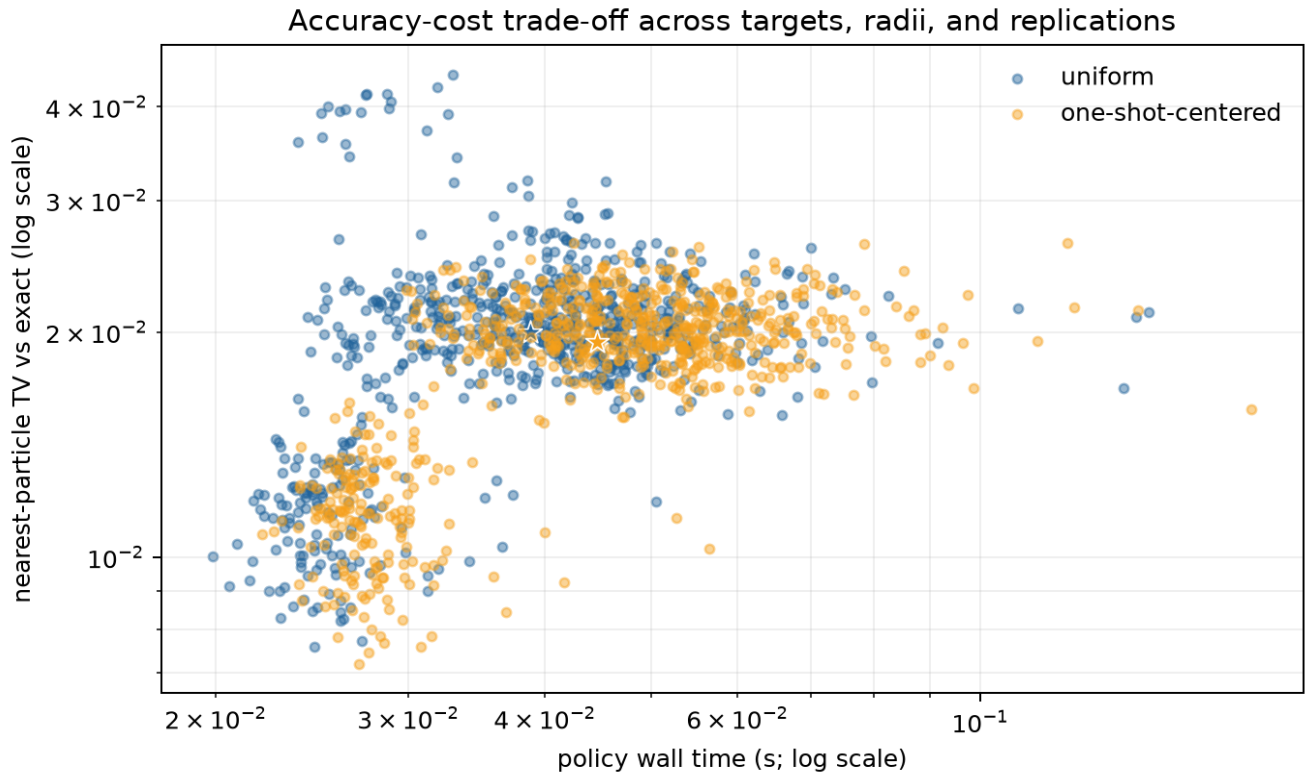
Policy	TV slope	BL slope
uniform	-0.03	-0.502
one-shot-centered	0.00108	-0.0447

These slopes are descriptive fits to the production medians, not theorem estimates. A slope closer to zero indicates slower decay of the measured finite-partition discrepancy over this radius range.

Representative angular diagnostics: lopsided comb, $\rho = 4$



2. Accuracy versus cost

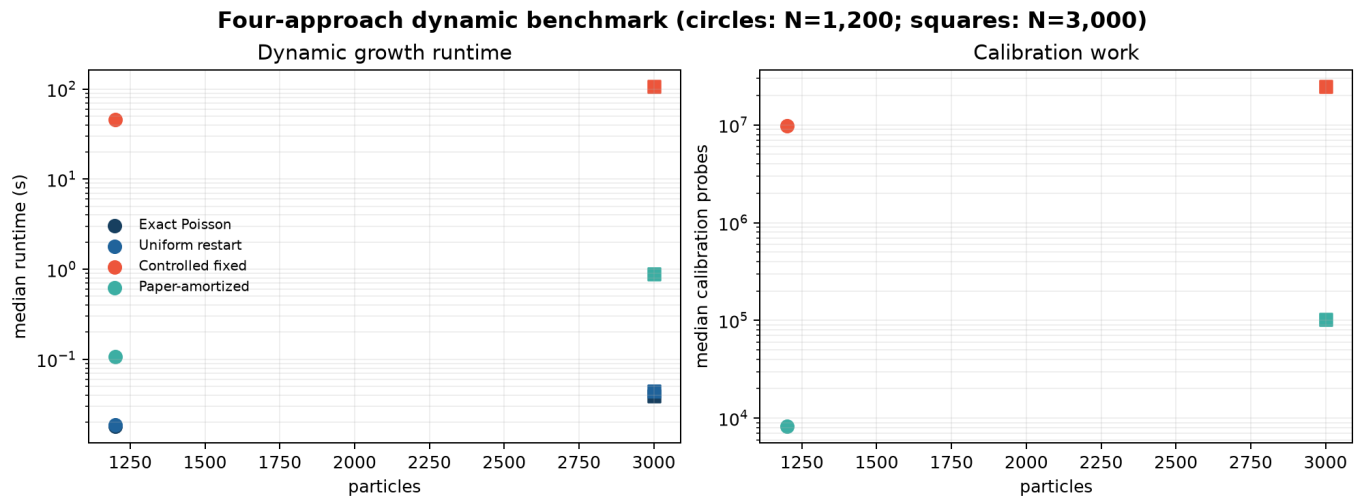


Each point is one target x radius x replication. Stars mark policy medians. The centered policy pays an independent search cost but changes the boundary center once, making it the relevant frozen-target approximation when the morphology is lopsided. Exact Poisson is deliberately not plotted as a competing policy here: it defines the reference attachment law against which the approximations are scored.

Production accounting

Policy	Median seconds	Median probes	Median TV
uniform	0.0388	5e+04	0.02
one-shot-centered	0.0446	5.82e+04	0.0194

3. Dynamic four-approach benchmark



N	Approach	Median runtime (s)	Median calibration probes	Speedup vs exact
1200	Exact Poisson	0.0181	0	1
1200	Uniform restart	0.0186	0	0.973
1200	Controlled fixed	45.8	9.82e+06	0.000395
1200	Paper-amortized	0.108	8.27e+03	0.168
3000	Exact Poisson	0.0396	0	1
3000	Uniform restart	0.0443	0	0.893
3000	Controlled fixed	106	2.46e+07	0.000374
3000	Paper-amortized	0.883	1.02e+05	0.0448

The dynamic run uses the package's four post-fix configurations at N = 1,200 and 3,000, seeds 42/137/911, and four Numba threads. Runtime differences reflect the intended scheduling trade-off: controlled-fixed spends substantial repeated calibration work, while paper-amortized spreads a certified schedule over growth. These runs compare work and certificates; three seeds are not evidence of morphology parity.

4. Reproducibility and limitations

Artifacts

The production manifest contains 720 shard records (4 targets x 9 radii x 20 replications), target digests, per-shard SHA-256 inventory, runtime provenance, deterministic batch seeds, geometry metadata, and the pilot decision. The pilot contains 180 records (4 x 9 x 5) and is revalidated before production. Dynamic outputs are the two benchmark JSON files consumed by this report.

What this demonstrates

The benchmark is a non-circular morphology stress test: it makes the target lopsided and increases the killing radius, exposing how a fixed center can retain finite-boundary bias while one-shot centering removes the leading geometric mismatch. It does not claim a theorem for non-circular death boundaries. The exact Poisson kernel remains optimal when exact return is available; the practical value of approach 4 is as a certified, lower-overhead approximation when exact return is unavailable or too costly.

Limitations

The static experiment uses circular death boundaries by design, finite nearest-particle and angular partitions, a fixed bounded-Lipschitz feature family, and only three stochastic DLA seeds. Wall times are machine-specific. The pilot's noise rule is mechanical and deliberately conservative; it triggered doubled evaluations for every tested radius in this run. Results should be read as validation of the predicted centering/error trend, not as a universal ranking for every geometry or workload.

Generated from the canonical pilot, production, and dynamic benchmark JSON artifacts. Report date: 2026-07-26.